

SCHWARTZ - ZIPPEL LEMMA

STATEMENT

P a polynomial of degree d

$$x_1, \dots, x_n \xleftarrow{\$} S$$

$$\mathbb{P}[P(x_1, \dots, x_n) = 0] \leq \frac{d}{|S|}$$

APPLICATION

☆ probabilistic test on polynomial identity

there is no a deterministic algorithm for polynomial identity testing

☆ bound on the total number of zeros

application on MQ-problem for cryptography

☆ primality testing

application for prime number test for cryptography

☆ matrix identity test

if $AB = CA$, the Sz-lemma give us a lower bound on the probability of $AB = CA$

☆ graph theory

CURIOSITY

the Schwartz-Zippel lemma real name is Schwartz-Zippel-DeMillo-Lipton lemma

There are several version of this lemma :

☆ Let Q a polynomial in variables $x_1, \dots, x_n$ and Q is not identically zero. Let d_i be the degree of x_i in Q and Q_i the coefficient of $x_i^{d_i}$ in Q . Inductively, let d_j be the degree of x_j in Q_j and $Q_{j,n}$ the coefficient of $x_j^{d_j}$ in Q_j . For $j=1, \dots, n$, let I_j be any set of elements of S . In the set $I_1 \times \dots \times I_n$, Q has at most $|I_1| \times \dots \times |I_n|$ zeros.

→ Corollary: $I_1 = \dots = I_n = S$, $|S| \geq c \deg(Q)$, in S^n , Q has at most $\frac{|S|^n}{c}$ zeros.

[Sch80]

☆ Let Q a polynomial in variables $x_1, \dots, x_n$ and Q is not identically zero. Let d_i be the degree of x_i in Q and Q_i the coefficient of $x_i^{d_i}$ in Q . Inductively, let d_j be the degree of x_j in Q_j and $Q_{j,n}$ the coefficient of $x_j^{d_j}$ in Q_j . Let $I_1, \dots, I_n = \{x \in R : x \leq L\}$ and L is the maximum value that Q assumes over I^n . Let d be a finite set of primes, and suppose that any product of any $m+1$ primes in d exceeds L . Then in $I_1 \times \dots \times I_n$, Q has at most $|I_1| \times \dots \times |I_n| \times \left(\frac{d_1}{|I_1|} \times \dots \times \frac{d_n}{|I_n|} \times \frac{m}{|d|} \right)$ points $(x_1, \dots, x_n, 0)$ such that $Q(x_1, \dots, x_n) = 0 \pmod p$.

→ Corollary: Let $|R| \geq c \deg(Q)$, $m = \frac{|d|}{c}$, hence Q has at most $\frac{2|d|^{n+1}}{c}$ points.

☆ Let Q a non-zero polynomial of total degree $d > 0$. Let $x_1, \dots, x_n \xleftarrow{\$} S$. Then $\mathbb{P}[P(x_1, \dots, x_n) = 0] \leq \frac{d}{|S|}$ and $|Z(P) \cap S^n| \leq d \cdot |S|^{n-1}$.

☆ Let Q a non-zero polynomial. Let D an integer such that the degree of x_i in Q is less D , $\forall i$, $\deg_{x_i}(Q) \leq D$. Let S an non empty set, then $|Z(P) \cap S^n| \leq |S|^n - (|S| - D)^n$ [GZ99]

[Sch80] — J.T.Schwartz — Fast Probabilistic Algorithms for Verification of Polynomials Identities
[Zip 79] — R. Zippel — Probabilistic algorithms for sparse polynomials